SURYA GS

Final-Year B.E. Electronics & Communication Engineering | VLSI / RTL / ASIC Design

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student specializing in VLSI and digital IC design, with hands-on exposure to Verilog RTL design, simulation, synthesis, FPGA development, Cadence EDA tools, and RTL-to-GDSII flow. Experienced through VLSI internship work in timing, placement, routing, and PPA analysis, with additional FPGA/TinyML design experience. Seeking entry-level or internship opportunities in RTL, ASIC, verification, FPGA, physical design, and semiconductor engineering.

TECHNICAL SKILLS

- HDL & Programming: Verilog, C, C++, Python.
- EDA & FPGA Tools: Cadence Xcelium, Genus, Innovus, Pegasus, Virtuoso; AMD Vivado; Quartus.
- VLSI Concepts: RTL Design, Simulation, Synthesis, RTL-to-GDSII, Static Timing Analysis (Basic), Placement & Routing, PPA Analysis.
- Circuit Knowledge: CMOS Device Physics, FinFET Fundamentals, Analog Circuit Fundamentals, Device Matching & Symmetry, Parasitic & RC Delay Analysis, DRC/LVS.
- Development Tools: VS Code, Google Colab, PyCharm.

INTERNSHIP EXPERIENCE

Digital Electronics & VLSI Intern - Codec Technologies Pvt. Ltd. | Nov 2025 - Feb 2026

- Worked on RTL design, simulation, and synthesis using Cadence EDA tools.
- Developed and verified digital modules with focus on timing and logic optimization.
- Gained exposure to backend implementation concepts including placement, routing, and PPA analysis.
- Built practical understanding of CMOS design principles and the VLSI implementation flow.

Embedded Systems Intern - Caliber Embedded Technologies India Pvt. Ltd. | Jun 2025

- Completed hands-on training in embedded system design and development, including microcontrollers, embedded programming, hardware interfacing, and real-time concepts.

SELECTED PROJECTS

Ultra-Low-Power Programmable Pacemaker ASIC – Ongoing

- Designing a configurable digital pacemaker ASIC architecture using Verilog RTL.
- Developing FSM-based control, sensing logic, pacing pulse generation, and programmable operating modes.
- Performing RTL simulation, logic synthesis, timing analysis, and physical-design exploration using Cadence EDA tools.

Real Time FPGA Tiny ML Speech Command Recognition | FPGA, TinyML, Vivado

- Designed a real-time speech command recognition system using a lightweight TinyML model deployed on FPGA hardware.
- Integrated speech preprocessing, feature extraction, and command classification for low latency edge inference.
- Demonstrated hardware acceleration for real-time embedded inference compared with software-only processing.

EDUCATION

B.E. Electronics and Communication Engineering, Mahendra Engineering College, Namakkal | 2022 - 2026 | CGPA: 7.47 / 10 (up to 6th semester)

- Higher Secondary (Class XII), Government Boys Higher Secondary School, Namagiripet | 2022 | 60%

ACHIEVEMENTS & CERTIFICATIONS

- Top 3% (39/1250) in AI/ML-Powered FPGA Design & Simulation Hackathon, Industry Academia Conclave 2025, NIT Jamshedpur with VLSI For All.
- Invited as Resource Person at Mahendra Engineering College for "High Performance System Design using EDA Tools" (9 Mar 2026), demonstrating Cadence-based VLSI workflows.
- INNOVAX-2025 Mini Project Expo - recognized for presenting "Fake News Detection".
- VLSI for Beginners -7Days Training Program, VLSI For All (Jan 2026).
- Semiconductor Circuit Design and Simulation ,VIT Vellore, sponsored by British Council (Jan 2026).
- Cadence SystemVerilog for Design and Verification v25.03 (2025); Cadence RTL-to-GDSII Flow v6.0 (2026); Cadence Basic Static Timing Analysis v3.0 (2026).